

Polynomial-time decision of local-unitary equivalence for graph states

Yuxuan Zhang

Institute of Physics, École Polytechnique Fédérale de Lausanne (EPFL)

CH-1015 Lausanne, Switzerland

Department of Physics, Princeton University

Princeton, New Jersey 08544, USA

22 September 2026

Abstract

Claudet and Perdrix gave a quasipolynomial-time algorithm for deciding local-unitary equivalence of labelled graph states. Its bottleneck is the computation of all edge changes induced by a generalized local complementation: the direct method imposes one incidence congruence for each of many vertex subsets. We show that the same congruences are generated by pair and triple rows under closure by fixed diagonal maps. Maintaining an integer Hermite basis makes this closure polynomial in both dimension and bit length. We then compute the full modular kernel and its binary image, preserving every admissible edge change. Substituting this computation into the Claudet–Perdrix reduction gives a deterministic polynomial-time algorithm for arbitrary finite simple graphs. The proof includes disconnected inputs and the case in which no generalized edge change is possible.

1 Introduction

Two quantum states are local-unitary equivalent when a change of basis on each individual subsystem transforms one into the other. For graph states, this asks whether a relation defined by arbitrary single-qubit unitaries can be recognized efficiently from two finite graphs.

A graph state assigns a qubit to each vertex of a finite simple graph. For a graph $G = (V, E)$, labelled on $V = [n]$, it is defined by

$$|G\rangle = \prod_{\{u,v\} \in E} CZ_{uv} |+\rangle^{\otimes n}, \quad |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}. \quad (1)$$

The local-unitary equivalence problem takes the binary adjacency matrices of two such graphs G and H as input and asks whether

$$|H\rangle = e^{i\phi} \left(\bigotimes_{v=1}^n U_v \right) |G\rangle \quad \text{for some } U_v \in U(2), \phi \in \mathbb{R}. \quad (2)$$

The answer is YES precisely when (2) holds. The equality is exact, and the vertex labels are fixed. A local complementation at v toggles every edge among distinct neighbors of v . We write $G \star v$ for the resulting graph and use LC for equivalence under sequences of these operations.

When the allowed unitaries are Clifford operators, local-unitary equivalence reduces to local complementation and can be decided in polynomial time [1, 2]. Arbitrary local unitaries give a

strictly larger relation: there are graph states that are LU-equivalent but not LC-equivalent [3]. The ordinary LC algorithm therefore does not decide the general problem.

Claudet and Perdrix obtained a graphical characterization of the larger relation [5] and an algorithm running in time $n^{\log_2 n + O(1)}$ [4]. Their algorithm first puts the graphs in a common standard form. It then computes all edge changes available through a generalized local complementation, and finishes with a constrained local-Clifford test. The expensive part is the middle computation: its direct formulation lists incidence congruences indexed by vertex subsets of logarithmically growing size. Whether the resulting quasipolynomial bound can be reduced to polynomial time is also asked explicitly in Claudet’s thesis [7, Chapter 8, p. 107].

Theorem 1.1 (Main result). *There is a deterministic classical algorithm that, given two finite simple graphs G and H on the same labelled vertex set of size $n \geq 1$, decides in polynomial bit complexity whether their graph states are local-unitary equivalent, up to global phase. No connectivity, degree, or graph-class promise is required.*

The proof replaces that middle computation. We first identify the precise object needed by the graph reduction: the complete binary space of admissible edge changes, together with multiplicities that realize its basis vectors. Computing a few valid changes, or deciding whether one particular change is possible, would not supply this object.

Our construction starts from the congruences defining admissible multiplicities. Although their rows are indexed by many subsets, adjoining a vertex to an index set applies one fixed diagonal map to the corresponding row. Pair and triple rows generate all the others under these maps. An integer lattice basis keeps this closure finite and small. Taking the annihilator of the resulting row module gives every admissible multiplicity; applying the edge-change map then gives the complete binary space required by the reduction.

Section 2 states the graph reduction in this form and derives the congruences. Section 3 constructs their row module without subset enumeration. Section 4 recovers the full kernel and binary image, after which Section 5 proves Theorem 1.1. The graph-reduction details are in Appendix A; the later appendices record exact computations, source locators, and the constrained Clifford test.

2 What the graph reduction requires

The purpose of the graph reduction is to replace the search over unitaries by one finite algebraic computation. We recall the needed results of Claudet and Perdrix, then specify that computation. Statement numbers for [4] refer to the arXiv v3 PDF; Appendix C gives the correspondence with its HTML rendering.

2.1 The surviving graph transformations

Connected components can be treated separately. LU-equivalence preserves the partition into labelled components, and two graphs with the same partition are equivalent exactly when the corresponding components are equivalent. Lemma A.1 proves this fact. We reject different partitions and accept one-vertex components, so consider a connected component of order $v \geq 2$.

Let LC_r denote local equivalence generated by single-qubit Hadamards and $Z(\pi/2^r)$ rotations, in the convention of [4, Section 2.3]; thus $\text{LC}_1 = \text{LC}$. Choose

$$r = \max\{1, \lceil \log_2(v+1) \rceil\}. \quad (3)$$

Then $v \leq 2^{r+3} - 1$, so [4, Corollary 29] identifies LU-equivalence with LC_r -equivalence on this component. In particular, $r = O(\log v)$ and $2^r \leq 2(v+1)$.

The common-standard-form algorithm [4, Lemmas 16–17] either rejects equivalence or transforms the two inputs by ordinary local complementations into representatives G_1, G_2 with a common partition

$$V = X \sqcup Z \sqcup W.$$

Here X, Z, W are the type- X , type- Z , and type- $\perp$ vertices relative to a common minimal-local-set cover. The full definition and preprocessing hypotheses are given in Appendix A.2. The properties needed now are that X is independent, $N(X) \subseteq Z$, and every vertex of X has the same neighborhood in both graphs. The preprocessing takes polynomial time, including the construction of the cover [6].

For such a surviving pair, [4, Lemma 18] reduces LC_r -equivalence to one generalized r -local complementation supported on X , together with ordinary local complementations at vertices of W . The generalized operation changes only edges within Z . The operations on W preserve all neighborhoods of X , so the two kinds of operations commute; the generalized operation may be performed first. This is the reason to compute all edge changes available on X .

2.2 From generalized complementation to congruences

For a multiset S of vertices, $s(x)$ denotes its nonnegative integer multiplicity and $\text{supp}(S) = \{x : s(x) > 0\}$. For $K \subseteq V$, let

$$c_S(K) = \sum_{x: K \subseteq N_G(x)} s(x) \quad (4)$$

be its common-neighbor multiplicity. The following conventions are those of [4, Definitions 1–2].

Definition 2.1. Let $r \geq 1$. An independent multiset S is r -incident if, for every $0 \leq j < r$ and every $K \subseteq V \setminus \text{supp}(S)$ with $|K| = j + 2$,

$$c_S(K) \equiv 0 \pmod{2^{r-j-\delta_{j0}}}. \quad (5)$$

Its r -local complementation toggles a pair outside $\text{supp}(S)$ precisely when its common-neighbor multiplicity is 2^{r-1} modulo 2^r . Pairs meeting the support are unchanged.

Every operation in Definition 2.1 is an LC_r transformation [5, 4]. Its input is a multiset: each vertex can participate with a multiplicity, and these multiplicities may be reduced modulo $q = 2^r$. Write $R = \mathbb{Z}/q\mathbb{Z}$ and $h = 2^{r-1}$. We now work with any graph G on n vertices and any prescribed independent set X ; the common standard form is needed only when returning to the full graph decision.

Put $Y = V \setminus X$, $m = |X|$, and $t = |Y|$. For each $v \in Y$, let $a_v \in \{0, 1\}^X$ record its neighbors in X . For $K \subseteq Y$, define

$$a_K = \bigodot_{v \in K} a_v, \quad w(k) = \begin{cases} 1, & k = 2, \\ k - 2, & k \geq 3. \end{cases} \quad (6)$$

The scalar product $a_K \cdot s$ counts the common neighbors of K , weighted by their multiplicities in s . For a pair, Definition 2.1 requires divisibility by 2^{r-1} , which is equivalent to $2a_K \cdot s = 0 \pmod{q}$. For $3 \leq k = |K| \leq r + 1$, it requires divisibility by 2^{r-k+2} , giving coefficient 2^{k-2} instead. Thus define

$$\Sigma = \{s \in R^X : 2^{w(|K|)} a_K \cdot s = 0 \pmod{q} \text{ for all } K \subseteq Y, 2 \leq |K| \leq r + 1\}. \quad (7)$$

In particular, pairs and triples have the same coefficient 2; the subsequent coefficients are 4, 8, $\dots$. This distinction will determine the starting rows of the compression algorithm.

Lemma 2.2. *The vectors in Σ , represented by multiplicities in $\{0, \dots, q-1\}$ on X , are exactly the r -incident multisets supported in X .*

Proof. For $K \subseteq Y$, the common-neighbor multiplicity is $a_K \cdot s$. When $k = |K|$, condition (5) requires divisibility by $2^{r-w(k)}$, exactly equivalent to (7). This equivalence also holds for $r = 1$, when the pair condition is vacuous.

The actual support of a multiplicity vector may be smaller than X . If a set $K \subseteq V \setminus \text{supp}(S)$ contains some $x \in X$, no vertex of X is adjacent to every vertex of K : independence of X forbids adjacency to x . Its common-neighbor multiplicity in S is therefore zero. All such extra conditions in Definition 2.1 hold automatically. These observations establish both directions. Reducing multiplicities modulo q preserves all relevant divisibilities. $\square$

2.3 The complete edge-change space

For every unordered pair $p \in \binom{Y}{2}$, define

$$f(s)_p = \frac{a_p \cdot \tilde{s}}{h} \pmod{2}, \quad s \in \Sigma, \quad (8)$$

where $\tilde{s}$ is any integer lift. Let $\Omega = f(\Sigma)$.

Lemma 2.3. *The map in (8) is well-defined and satisfies*

$$f(s+t) = f(s) + f(t), \quad f(cs) = (c \bmod 2)f(s). \quad (9)$$

Consequently Ω is a binary linear subspace.

Proof. The pair constraint is $2a_p \cdot s = 0 \pmod{q}$, so every integer lift has $a_p \cdot \tilde{s}$ divisible by h . For $r = 1$, this is automatic because $h = 1$. Changing the lift by qz changes the quotient by $2a_p \cdot z$, an even integer. The map is therefore well-defined.

If $\widetilde{s+t} = \tilde{s} + \tilde{t} - qz$, then

$$\frac{a_p \cdot \widetilde{s+t}}{h} = \frac{a_p \cdot \tilde{s}}{h} + \frac{a_p \cdot \tilde{t}}{h} - 2a_p \cdot z. \quad (10)$$

Reducing modulo two removes the last term. For a lift $\tilde{c}$ of a scalar $c \in R$, writing $\tilde{c}s = \tilde{c}\tilde{s} - qz$ gives the scalar identity in the same way. Reduction $R \rightarrow \mathbb{F}_2$ is well-defined because q is even, establishing (9). $\square$

For a multiplicity vector in Σ , the coordinates of $f(s)$ specify exactly which pairs are toggled by its generalized complementation. Thus $\Omega = f(\Sigma)$ describes all possible edge changes. In a common standard form, its coordinates outside $\binom{Z}{2}$ vanish because vertices of W have no neighbors in X .

Proposition 2.4 (Reduction to the edge-change space). *Suppose G_1, G_2 are a surviving connected pair in common standard form, with the same types and the same neighborhoods on X . Given a binary basis $\omega_1, \dots, \omega_d$ of the complete space Ω for G_1 and vectors $s_i \in \Sigma$ with $f(s_i) = \omega_i$, their LC_r -equivalence can be decided in time polynomial in the graph order, r , and the supplied data. The decision also applies when $d = 0$.*

This is the algorithmic use of [4, Lemmas 20–21]. A basis vector is implemented by a group of degree-two auxiliary vertices. Selecting that group performs the corresponding edge change, and homogeneous constraints enforce that groups are selected in their entirety. The constrained LC algorithm decides whether any such combination, followed by operations on W , reaches

G_2 . Appendix A gives the proof, checks the connectedness and even-degree hypotheses of the constrained test, and handles $d = 0$ separately.

Proposition 2.4 identifies the remaining task: compute a basis of Ω and its witnessing multiplicities. In [4, Lemma 19], the construction writes down every row of (7), for sets of sizes up to $r + 1$. The number of possible index sets gives the quasipolynomial cost when r grows logarithmically with the graph order. We will obtain the same Σ and the same Ω while replacing this row enumeration.

3 Compressing the incidence constraints

Equation (7) asks that s annihilate every weighted incidence row. It is enough to find a small generating set for their row module: it has exactly the same annihilator. Index Y by $[t]$, keeping the vectors and weights of (6), and let

$$M_r = \text{span}_R\{2^{w(|K|)}a_K : 2 \leq |K| \leq r + 1\} \subseteq R^m. \quad (11)$$

Subsets larger than t contribute no rows. We construct generators of M_r without enumerating the higher-order subsets. The calculation works for arbitrary binary vectors $a_1, \dots, a_t \in \{0, 1\}^m$. Throughout, complexity is measured in bit operations and is polynomial in the value of r ; Section 2 has already fixed a logarithmic value for the graph application.

Theorem 3.1 (Polynomial incidence compression). *A generating matrix for M_r with at most m rows can be constructed deterministically in time polynomial in m, t, r . The construction uses at most*

$$B(r, t) = \max\{0, \min\{r - 2, t - 3\}\} \quad (12)$$

strict lattice enlargements, and its retained integer basis entries have $O(r + 1)$ bits.

3.1 Why pairs and triples suffice

For $|K| \geq 3$ and $v \notin K$, the next row $2^{|K|-1}a_{K \cup \{v\}}$ is obtained from $2^{|K|-2}a_K$ by doubling and coordinatewise multiplication by a_v . This gives the R -linear maps

$$T_v(z) = 2a_v \odot z. \quad (13)$$

Let N_r be the smallest submodule containing $2a_K$ for $|K| = 2$ or 3 and invariant under all the maps T_v .

Lemma 3.2. $M_r = N_r$.

Proof. The pair seeds occur in (11). The triple seeds occur when $r \geq 2$; when $r = 1$, all seeds are zero modulo two. Thus every seed belongs to M_r .

We check invariance on a generating row $2^{w(k)}a_K$, where $k = |K|$. If $v \in K$, the binary coordinates satisfy $a_v \odot a_K = a_K$, so T_v gives twice that row. If $v \notin K$ and $k = 2$, the result is $4a_{K \cup \{v\}}$, twice the triple seed. For $r = 1$ this result is zero. If $v \notin K$ and $k \geq 3$, then

$$T_v(2^{k-2}a_K) = 2^{k-1}a_{K \cup \{v\}}, \quad (14)$$

which is the next prescribed row, unless $k = r + 1$, in which case it vanishes modulo 2^r . Hence M_r is invariant and $N_r \subseteq M_r$.

Conversely, if $|K| = k \geq 4$, choose three vertices of K and start with their triple seed. Applying T_v for the remaining $k - 3$ distinct vertices gives coefficient $2 \cdot 2^{k-3} = 2^{k-2}$. Every generator in (11) is therefore in N_r . Pair and triple generators were already seeds. This proves equality; in particular, repeated application of a vertex introduces no extraneous constraints. $\square$

3.2 A bounded lattice computation

We fix the following row-Hermite convention. For a full-rank lattice $L \subseteq \mathbb{Z}^m$, its basis matrix H is lower triangular, with

$$H_{ii} > 0, \quad H_{ij} = 0 \ (j > i), \quad 0 \leq H_{ij} < H_{jj} \ (j < i). \quad (15)$$

The rows generate L over $\mathbb{Z}$. Transposing a column-Hermite computation gives this convention.

Lemma 3.3 (Size and lifting of a lattice basis). *Suppose $q\mathbb{Z}^m \subseteq L \subseteq \mathbb{Z}^m$, and let H be its row-Hermite basis. Each H_{ii} divides q , and every entry of H lies between zero and q . If $\pi : \mathbb{Z}^m \rightarrow (\mathbb{Z}/q\mathbb{Z})^m$ is reduction modulo q , then $L = \pi^{-1}(\pi(L))$.*

Proof. For each i , write $qe_i = cH$ with $c \in \mathbb{Z}^m$, since $qe_i \in L$. Compare columns from m down to $i + 1$. Lower triangularity and the nonzero pivots successively give $c_j = 0$ for all $j > i$. Column i then gives $c_i H_{ii} = q$, proving $H_{ii} \mid q$. The off-diagonal bounds follow from (15).

If $z \in \pi^{-1}(\pi(L))$, choose $y \in L$ with $z - y \in q\mathbb{Z}^m$. Both y and $z - y$ lie in L , so $z \in L$. The other inclusion is immediate. In particular, two lattices containing $q\mathbb{Z}^m$ are equal whenever their images modulo q are equal. $\square$

Proof of Theorem 3.1. The case $m = 0$ is immediate, so assume $m > 0$. Lift the seed rows to integers and let $L_0 \subseteq \mathbb{Z}^m$ be the lattice generated by those rows and $q\mathbb{Z}^m$. Extend (13) to the same integer diagonal map and iterate

$$L_{i+1} = L_i + \sum_{v=1}^t T_v(L_i). \quad (16)$$

After each step, retain a full-rank row Hermite basis. Applying T_v to a basis suffices to generate $T_v(L_i)$.

The image of every L_i modulo q is contained in M_r , by the invariance proved in Lemma 3.2. Conversely, a prescribed row indexed by a set of size $k \geq 4$ is obtained from a triple seed after $k - 3$ applications of the maps. The largest relevant size is $\min\{t, r + 1\}$. Thus every prescribed row is present by step $B(r, t)$; when $r \leq 2$ or $t \leq 3$, the seeds already suffice. Every L_i contains $q\mathbb{Z}^m$, and $T_v(q\mathbb{Z}^m) \subseteq q\mathbb{Z}^m$. Lemma 3.3 therefore makes equality of their modular images equivalent to equality of the integer lattices. Hence $L_{B(r,t)}$ is stable. If equality occurs earlier, invariance prevents any later change. A further Hermite computation detects stability.

The initial matrix has at most $m + \binom{t}{2} + \binom{t}{3}$ rows and m columns. Every subsequent matrix has at most $m(1 + t)$ rows. By Lemma 3.3, every retained entry lies in $[0, q]$ and has $O(r + 1)$ bits. Applying T_v gives entries of magnitude at most $2q$. There are at most $B(r, t) + 2$ Hermite computations: the initial reduction, at most $B(r, t)$ strict enlargements, and a final stability check.

Deterministic Hermite-normal-form algorithms have polynomial bit complexity in their matrix dimensions and input bit lengths [8, Theorem 2 and the rectangular extension $H(m, n, A)$, p. 504]. Apply that full-row-rank rectangular algorithm to the transpose of our generating matrix, which has rank m because it contains qI_m . Polynomially many calls on the bounded matrices above establish the asserted bit complexity, including intermediate arithmetic inside each call. Reducing the final m basis rows modulo q gives a generating matrix B for M_r . Zero rows may be omitted. These are module generators; they need not form a free basis over R . $\square$

Algorithm 1: weighted incidence compression

1. Form the pair and triple rows $2a_K$ and the rows of qI_m .
2. Replace their integer row span by a canonical Hermite basis H .
3. Compute a Hermite basis H' of the rows of H and all $T_v(H)$.
4. If $H' = H$, return $H \bmod q$; otherwise set $H \leftarrow H'$ and repeat step 3.

For $m = 0$, return the empty matrix immediately.

Corollary 3.4. *For $s \in R^m$, the exponentially indexed conditions*

$$2^{w(|K|)} a_K \cdot s = 0 \pmod{q} \quad (2 \leq |K| \leq r+1) \quad (17)$$

hold if and only if $Bs = 0$ in R .

Proof. Annihilating generators of a row module is equivalent to annihilating the entire module. Apply Theorem 3.1. $\square$

The graph problem has therefore been reduced to a matrix with at most m rows. Corollary 3.4 preserves the entire admissible set Σ , which is what allows the next step to recover all of Ω .

4 Recovering the complete edge-change space

Let B be the generating matrix from Theorem 3.1. Corollary 3.4 gives

$$\Sigma = \ker_R B. \quad (18)$$

To finish the computation required by Proposition 2.4, we need generators of this entire kernel and a basis of its image under f . This gives the following result.

Theorem 4.1 (Complete image with preimages). *In polynomial bit complexity in n, r , one can compute a basis $\omega_1, \dots, \omega_d$ of Ω , with $d \leq |X|$, and vectors $s_i \in \Sigma$ satisfying $f(s_i) = \omega_i$. Membership in Ω is decidable in polynomial time, with a witnessing multiplicity vector for every member.*

4.1 Recovering every admissible multiplicity

For $r > 1$, R is not a field, so field Gaussian elimination cannot be used to assert that all solutions have been recovered. The integer lattice from the compression gives a direct way to generate the kernel, including its nonfree part.

Lemma 4.2. *Given a $b \times m$ matrix B over R , with $b \leq m$, a generating set of $\ker_R B$ with at most m elements can be computed deterministically in polynomial bit complexity in m, r .*

Proof. The case $m = 0$ is immediate. Otherwise, let H be a full-rank row Hermite basis of the integer lattice generated by a lift of B and qI_m . The conditions $Bs = 0 \pmod{q}$ and $HS = 0 \pmod{q}$ are equivalent. Because the row lattice of H contains $q\mathbb{Z}^m$, there is an integer matrix C with $qI_m = CH$. It follows that

$$K = qH^{-1} = C \quad (19)$$

is an integer matrix.

The columns of K , reduced modulo q , generate the kernel. Each column is a solution because $HK = qI_m$. Conversely, for any integer lift of a solution s , write $HS = qz$ with $z \in \mathbb{Z}^m$. Then

$s = Kz$, proving completeness. The generators need not be independent over R ; no freeness assumption is made.

Computing H has polynomial bit complexity, and Lemma 3.3 bounds its entries by q . There is no need to compute a rational inverse. Solve $HK = qI_m$ by forward substitution, column by column:

$$k_{ij} = \frac{q\mathbf{1}_{i=j} - \sum_{\ell < i} H_{i\ell} k_{\ell j}}{H_{ii}}, \quad i = 1, \dots, m. \quad (20)$$

Every division is exact, because (19) already proves that the unique solution is integral. For a fixed column j , the entries above j vanish and $|k_{jj}| \leq q$. If $S_i = \sum_{\ell \leq i} |k_{\ell j}|$, then $S_i \leq (1+q)S_{i-1}$ for $i > j$. Hence

$$|k_{ij}| \leq q(1+q)^{m-1}. \quad (21)$$

All entries and intermediate sums have $O(m(r+1) + \log(m+1))$ bits. Equation (20) uses $O(m^3)$ integer arithmetic operations on numbers of this size, so even elementary integer algorithms suffice for polynomial bit complexity. Finally reduce the columns modulo q . If B came from Algorithm 1, reuse its already computed integer basis H .

For example, $B = (2)$ over $\mathbb{Z}/4\mathbb{Z}$ gives $H = (2)$ and $K = (2)$. Its kernel is $\{0, 2\}$, generated by an element annihilated by two. This illustrates why treating R as a field, or assuming a free kernel, would be incorrect. $\square$

4.2 Taking the binary image and retaining witnesses

The map f is a homomorphism to a binary vector space by Lemma 2.3. Applying it to a full kernel generating set therefore gives all possible edge changes; retaining the preimages makes each resulting basis vector realizable.

Proof of Theorem 4.1. Apply Lemma 4.2 to obtain generators $g_1, \dots, g_\ell$ of Σ , with $\ell \leq m$. For each g_j and pair p , compute the residue $c_p = a_p \cdot g_j \bmod q$ in $\{0, \dots, q-1\}$. It belongs to $\{0, h\}$ by the pair constraint. Thus $f(g_j)_p = c_p/h$ is an exact integer division; it does not divide by a nonunit in R .

Every $s \in \Sigma$ is an R -linear combination $\sum_j c_j g_j$. By Lemma 2.3,

$$f(s) = \sum_j (c_j \bmod 2) f(g_j). \quad (22)$$

Conversely, every binary combination of the images is the image of the corresponding sum of the generators. Hence these at most m images span all of Ω .

Perform binary Gaussian elimination on them, carrying along their preimages over R . Whenever image rows are added, add the corresponding preimages modulo q . The invariant $f(s_i) = \omega_i$ survives by (9). Remove dependent zero image rows. If $\delta \in \Omega$, solve $\delta = \sum_i \epsilon_i \omega_i$ over $\mathbb{F}_2$ and return $s = \sum_i \epsilon_i s_i \bmod q$.

There are $\binom{|Y|}{2} = O(n^2)$ image coordinates and at most m generators. Image computation uses $O(n^2 m^2)$ elementary arithmetic operations. Before reducing a dot product modulo q , its value is at most $m(q-1)$, requiring $O(r + \log(m+1))$ bits. Binary elimination uses $O(n^2 m^2)$ field operations, and at most $O(m^3)$ additions of r -bit witness coordinates. Thus the carried witnesses do not hide a growing precision requirement. Combined with the compression and kernel bounds, this proves polynomial bit complexity. An equation description of Ω may alternatively be obtained from a basis of its binary orthogonal complement. $\square$

Corollary 4.3 (One prescribed-support operation). *A graph H is obtained from G by one valid r -local complementation supported in X if and only if the two graphs agree on pairs meeting*

X and their edge difference on Y belongs to Ω . The condition is decidable in polynomial bit complexity in n, r , and an accepting instance has constructible binary-encoded multiplicities.

Proof. Lemma 2.2 identifies all valid multiplicities. For a pair in Y , the toggle predicate is exactly (8). A pair meeting the actual support is unchanged by definition. A pair meeting X outside that support has no common neighbor in X , so its common-neighbor multiplicity is zero and it is unchanged as well. These facts prove necessity and sufficiency. Representatives below q have at most r bits each. $\square$

5 Proof of the main theorem

The two required results are now available. Proposition 2.4 reduces each surviving graph pair to the complete edge-change space. Theorem 4.1 computes that space, with witnesses, in polynomial bit complexity. Their composition gives the following algorithm.

Algorithm 2: labelled graph-state LU equivalence

1. Compare labelled connected-component partitions. Reject if they differ.
2. For each common component of order $v \geq 2$, choose r by (3).
3. Use [4, Lemmas 16–17] to standardize and check types and type- X neighborhoods. Reject on failure.
4. Use Algorithm 1 and Theorem 4.1 to compute the complete toggle basis.
5. If $d = 0$, decide ordinary LC on the original standard forms. If $d > 0$, build the matched gadgets and decide constrained LC using (29).
6. Accept exactly when every component test accepts; singletons always accept.

Proof of Theorem 1.1. Lemma A.1 justifies the connected-component test. The standard-form procedure and the type and neighborhood comparisons in Section 2 reject only inequivalent inputs. On every surviving component, the choice (3) makes LU-equivalence equivalent to LC_r -equivalence. Theorem 4.1 supplies precisely the full space and the multiplicities assumed by Proposition 2.4, whose decision is therefore both sound and complete. This includes the zero-dimensional image. All operations preserve vertex labels, and tensoring the component transformations proves the equivalence for the original graphs.

It remains to bound the cost. On a component of order v , preprocessing takes $O(v^{6.38})$ time. Theorems 3.1 and 4.1 take polynomial bit complexity in v, r ; the chosen value satisfies $r = O(\log v)$ and $q \leq 2(v + 1)$. These algebraic computations therefore have polynomial bit complexity in v .

When $d > 0$, the auxiliary construction in Appendix A uses at most $d \binom{v}{2}$ new vertices. The bound $d \leq |X|$ from Theorem 4.1 gives

$$N \leq v + d \binom{v}{2} \leq v + |X| \binom{v}{2} = O(v^3). \quad (23)$$

The constrained test has at most $2N$ additional linear equations, so Lemma D.1 bounds this stage by $O(N^4) = O(v^{12})$ binary operations. This exponent concerns the graph decision stage; no optimized exponent for the separate integer-lattice computations is asserted. When $d = 0$, the algorithm uses the ordinary polynomial LC test on the original standard forms. Summing a fixed polynomial bound over the components remains polynomial in n . $\square$

6 Concluding remarks

The resulting algorithm has polynomial worst-case complexity, but its practical cost remains to be established. The auxiliary graph can have cubic size, and the current diagnostic code does not include the polynomial MLS-cover and standardization routines. A complete implementation would allow comparison with direct subset enumeration on small instances and would identify which part of the reduction dominates in practice.

Code and preparation

The [companion page](#) provides the source, exact checking scripts, recorded outputs, and the original proof submitted in [QIQC issue 97](#). The computations and their coverage are described in [Appendix B](#); licensing and revision records are included in the archive.

AI assistance was used to develop the arguments, write the diagnostic code, and prepare the manuscript under the author’s direction. Scientific Agent Skills [9] supplied scientific-writing and evidence-tracking guidance. Independent expert review and historical-priority verification are pending.

A Details of the graph reduction

We prove Proposition 2.4 using the graph reductions of [4]. The required input is the complete space Ω and its supplied preimages. None of the arguments in this appendix requires an algorithm for computing that input.

A.1 Connected components

Lemma A.1. *LU-equivalent labelled graph states have the same partition of their vertex set into connected components. Equivalence then holds if and only if it holds on each corresponding component.*

Proof. A graph state factors across a labelled cut $A : V \setminus A$ if and only if no edge crosses that cut. To see the nontrivial direction, remove the diagonal phases contributed by edges lying entirely within either side of the cut. Up to a common normalization, the bipartite coefficient matrix becomes

$$C_{x,y} = (-1)^{x^\top \Gamma_{A,V \setminus A} y}. \quad (24)$$

If a crossing entry of Γ equals one, the rows indexed by the zero vector and the corresponding single-bit vector, and the analogous two columns, give the minor

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (25)$$

It has nonzero determinant. The state therefore has Schmidt rank at least two across the cut. With no crossing edges, the coefficient matrix factors.

Product unitaries preserve factorization across every fixed labelled cut. The finest partition into graph-state product factors is thus the connected-component partition, which must be the same for both inputs. When partitions coincide, a global product-unitary equality implies equality of the pure-state factors up to component phases, for example by taking reduced states on each component. Conversely, tensoring component transformations supplies a transformation on the whole graph, with their phases absorbed into the global phase. $\square$

A.2 Standard form

Write Γ for the adjacency matrix over $\mathbb{F}_2$, and X_u, Z_u for single-qubit Paulis. The graph-state stabilizer has generators $X_u \prod_{v \in N(u)} Z_v$. For a vertex set D with indicator d , their product has support $D \cup \text{Odd}_G(D)$, where $\text{Odd}_G(D)$ is the set of vertices with an odd number of neighbors in D . At vertex u its Pauli, up to phase, is specified by $(d_u, (\Gamma d)_u)$. The three nonzero pairs $(1, 0), (1, 1), (0, 1)$ represent X, Y, Z , respectively.

A local set is a nonempty support of this form. An MLS cover $\mathcal{M}$ is a family of inclusion-minimal local sets covering every vertex. For a vertex u , consider every generator D of every member of $\mathcal{M}$ containing u . Its type is X, Y , or Z if all these Pauli pairs equal that same nonzero pair, and $\perp$ otherwise [4, Definitions 12–13, pp. 8–9]. A cover can be computed in polynomial time [6]; the algorithm does not enumerate all minimal local sets.

Standard form has three requirements [4, Definition 14, p. 9]: no vertex has type Y ; every neighbor v of a type- X vertex u has type Z and satisfies $u \prec v$ in the fixed vertex order; and each star $\{u\} \cup N(u)$ at a type- X vertex belongs to $\mathcal{M}$. In particular, the type- X set is independent.

The common-standard-form procedure [4, Lemma 16 and Appendix C] either rejects LU-equivalence soundly or returns ordinary-LC-equivalent representatives in standard form with respect to a common MLS cover, in $O(n^{6.38})$ time. For LU-equivalent representatives, corresponding types and type- X neighborhoods agree [4, Lemma 17]. The characterization of generalized complementation reduces the remaining problem to a single operation: surviving standard forms are LC_r -equivalent if and only if one r -local complementation supported on their type- X set, together with ordinary complementations on type- $\perp$ vertices, transforms one into the other [4, Lemma 18]. The reverse implication follows because each permitted operation is an LC_r operation.

Write X, Z, W for the type- X , type- Z , and type- $\perp$ sets. Standard form gives $N(X) \subseteq Z$. Ordinary complementation at $w \in W$ changes only edges among $N(w)$, which is disjoint from X . It therefore preserves all neighborhoods of X . A generalized operation on X changes only Z – Z edges, preserving $N(w)$. Each operation thus preserves the data specifying the other's toggles, so they commute. The same reasoning applies throughout a sequence on W : no such step creates an edge from W to X .

Let G_1, G_2 be a surviving connected pair and suppose the data in Proposition 2.4 have been supplied. Ordinary local complementation preserves connectedness: any removed edge is replaced by a two-edge path through the complemented vertex. Thus both standard forms are connected. The only nonzero coordinates of Ω are pairs in Z , because vertices outside $X \cup Z$ have no neighbor in X . We use these coordinates in the auxiliary construction below.

A.3 Replacing edge changes by local complementations

Suppose $d = \dim \Omega > 0$. A degree-two vertex provides a simple way to toggle a chosen edge: complementing the vertex toggles the edge between its two neighbors. We use this construction for each nonzero coordinate of each basis vector, as in [4, Lemma 20]. For each basis vector ω_i and each pair $\{u, w\} \subseteq Z$ on which it is one, add a new vertex p_{uw}^i joined only to u, w , after deleting X . Use the same new labels in both graphs. Let P_i be the set of new vertices for ω_i . The resulting graph order satisfies $N \leq v + d \binom{v}{2}$. Since a binary basis has $d \leq \binom{|Z|}{2}$, this construction has polynomial size.

Lemma A.2 (Applicability of constrained LC). *When $d > 0$, both augmented graphs are connected and have an even-degree vertex.*

Proof. Every basis vector is nonzero, so at least one auxiliary vertex is added; each such vertex

has degree two.

For connectivity, fix $x \in X$. The multiplicity vector $h\mathbf{e}_x$ is in Σ , because every weighted incidence row has an even coefficient and hence its scalar product with this vector is divisible by q . Its toggle vector is exactly the clique on $N_{G_1}(x)$. As the basis vectors span Ω , every pair of distinct neighbors of x must occur in at least one basis vector. Some auxiliary vertex therefore supplies a path of length two between each such pair.

Take any two vertices outside X and a simple path between them in G_1 . Independence of X ensures that each passage through an internal vertex $x \in X$ has two neighbors outside X . Replace that passage by the auxiliary path just described. Vertices of X with degree at most one cannot occur internally on the original simple path. Thus the remainder together with the auxiliary vertices is connected. The remainder is nonempty because G_1 is connected, has at least two vertices, and X is independent. All new vertices attach to that remainder. The same argument applies to G_2 , using its connectedness and the identical neighborhoods on X . $\square$

Lemma A.3 (Restricted graph operations). *The original standard forms are LC_r -equivalent if and only if their augmented graphs are related by ordinary local complementations on $W \cup \bigcup_i P_i$, with every group P_i either used in its entirety, once per vertex, or not used.*

Proof. Following [4, Lemma 20 and Appendix D], first observe that the auxiliary vertices form an independent set, have neighbors only in Z , and have no neighbors in W . Complementing p_{uw}^i toggles only the edge $\{u, w\}$. Using all of P_i therefore adds ω_i to the edge vector on Z .

A complementation at $w \in W$ preserves every auxiliary neighborhood, since none contains w . Complementation at an auxiliary vertex preserves $N(w)$, since it changes only Z – Z edges. The operations commute, and these properties persist after each allowed step. Thus an allowed sequence can be ordered with all auxiliary operations first, followed by operations on W .

If the standard forms are LC_r -equivalent, their transformation has a toggle $\omega \in \Omega$ followed by operations on W . Express $\omega = \sum_i \epsilon_i \omega_i$ and use exactly the groups with $\epsilon_i = 1$. After deleting X , this performs the same toggle; the remaining operations on W have the same effect. Conversely, an allowed augmented sequence selects a binary sum $\omega \in \Omega$. The supplied preimages give the valid multiplicity vector $s = \sum_i \epsilon_i s_i \bmod q$ with this image, by Lemma 2.3. Restore X , perform its generalized operation, and then the same sequence on W . The original neighborhoods of X are unchanged throughout, so the result is exactly the second standard form. $\square$

The restrictions in Lemma A.3 become $\ell \leq 2N$ homogeneous linear equations in the binary local-Clifford coordinates [4, Lemma 21]. On connected graphs with an even-degree vertex, the constrained LC algorithm solves the resulting problem in $O((N^2 + \ell)N^2)$ binary operations [4, Proposition 10]. Lemma A.2 supplies both hypotheses. Appendix D gives the equations, their conversion to the restricted graph operations, and the determinant test used by this algorithm.

A.4 When no generalized edge change is possible

Lemma A.4. *If $\Omega = \{0\}$, then G_1 and G_2 are LC_r -equivalent if and only if they are ordinarily LC-equivalent.*

Proof. One implication is immediate. For the other, [4, Lemma 18] expresses a LC_r equivalence by ordinary complementations on W and one valid generalized operation on X . These operations commute. Since $\Omega = \{0\}$, the latter operation changes no edge of G_1 and can be omitted. Only ordinary complementations remain. $\square$

In this case run the ordinary LC algorithm [2] on the original standard forms G_1, G_2 . Deleting X first could leave disconnected graphs or graphs without an even-degree vertex, outside the constrained routine’s hypotheses. Lemma A.4 supplies a separate exact decision rule, including the case $X = \emptyset$.

Proof of Proposition 2.4. For $d > 0$, Lemma A.3 identifies LC_r -equivalence with the restricted operations on the two auxiliary graphs. Lemma A.2 gives the hypotheses needed by the constrained LC test in Appendix D. The graphs have polynomial order in the input size and d , and the constrained test has polynomial complexity in that order. For $d = 0$, Lemma A.4 gives the ordinary LC test on the original standard forms. Both branches decide exactly the required equivalence in polynomial time. $\square$

B Exact computations

The accompanying scripts compare the algebraic constructions with direct enumeration and test the restricted graph reduction on finite examples. All computations use exact arithmetic. They check individual parts of the construction; the polynomial MLS standardization algorithm is not implemented. The proof of Theorem 1.1 is the argument in the main text.

B.1 Algebraic checks

The first comparisons cover 45 closure systems and 36 kernel-and-image systems; all agree with direct enumeration. The closure examples all stabilize at the seed lattice, so they do not test a strict enlargement. To exercise this step, a further family uses all Boolean column patterns on $t \in \{3, 4, 5, 6\}$ rows, at six levels. Twelve of its 24 systems require strict enlargement; retaining the seeds alone fails in each of those twelve. The script also checks all 2×2 matrices over $\mathbb{Z}/4\mathbb{Z}$, 384 binary incidence systems, and five boundary cases. The scripts and outputs are included in the archive.

A second kernel implementation uses the dual-lattice formula (19); the earlier implementation uses modular row and column elimination. It checks every binary 3×3 incidence matrix at levels two and three. It compares the full kernel and image with the literal incidence definition, including the actual support of each multiplicity vector, and verifies every binary combination of the retained image-basis witnesses. This covers 1,024 systems, 294,912 multiplicity vectors, and 2,928 carried witnesses. The dual-lattice method also reproduces every kernel of a 2×2 matrix over $\mathbb{Z}/4\mathbb{Z}$, including 81 kernels of cardinality two or eight, which are nonfree over that ring.

Sixteen additional Boolean-pattern systems test the closure bound at levels up to $r = 256$. All agree with direct subset enumeration, and all attain the bound $B(r, t)$ for their parameters. Two deliberately incorrect variants—omitting triple seeds and omitting the factor two in T_v —give different lattices from the exact construction on the designated control instances.

A focused check of the forward-substitution formula (20) revisits the 256 general mod-four matrices and these sixteen Boolean-pattern systems, together with three unconstrained boundary cases. Every division is exact, all resulting matrices agree with qH^{-1} , and all exhaustively enumerated mod-four kernels agree. It reuses the preceding script’s Hermite and incidence routines and therefore checks the recurrence within the same implementation.

B.2 Checks of the graph reduction

For connected labelled graphs of orders two through four, we enumerate independent supports and levels one through three. When the toggle space is nonzero, we compare two computations:

direct enumeration of the possible toggles followed by graph search over the allowed ordinary complementations, and the auxiliary-graph construction followed by the constrained local-Clifford equations. There are 2,460 comparisons: 1,020 accept and 1,440 reject, with no disagreements. In each case we also enumerate the entire linear solution space of the Clifford equations before checking the determinant conditions. These tests exercise the restricted graph-operation reduction; arbitrary chosen supports need not be type- X sets of a standard form. The 261 zero-image support/level cases are therefore excluded from this comparison, since Lemma A.4 relies on standard form. A further 1,105 ordinary-LC comparisons agree with unrestricted graph search on small connected graphs satisfying the even-degree hypothesis.

We also use the 27-vertex construction of Tsimakuridze and Gühne [3, Section 7 and Figure 7] to test an operation beyond ordinary LC. Let Z contain six vertices and let X contain one vertex for each four-element or five-element subset of Z , adjacent precisely to that subset. Thus $|X| = 21$. Let G be this bipartite graph, and let H add the clique on Z . Giving every vertex of X multiplicity one yields ten common neighbors for every pair in Z and six for every triple. At level two, all incidence conditions hold and every pair is toggled. Hence this operation takes G to H .

The stars at vertices of X form a common MLS cover for which both graphs are in standard form. To check this, for each of the 42 stars in the two graphs the code solves for stabilizers supported inside the star and finds only its defining nonidentity generator. The computed toggle space has dimension 15 and contains the clique, with an explicitly recovered multiplicity witness. The constrained gadget test accepts. In contrast, enumerating all 2^8 solutions of the ordinary-LC linear equations finds none satisfying the local determinant conditions, so the ordinary-LC test rejects. The graph pair and its LU-but-not-LC property are from [3]; our computations use this known example as a diagnostic control for the reduction.

Test family	Cases	Outcome
Boolean-pattern systems, $r \leq 256$	16	Exact closure and bound
Dual kernels of 2×2 matrices over $\mathbb{Z}/4\mathbb{Z}$	256	All kernels agree
Binary 3×3 incidence systems	1,024	Full kernels and images agree
Carried image-witness combinations	2,928	All valid
Constrained gadget comparisons	2,460	1,020 yes; 1,440 no
Ordinary-LC graph-search comparisons	1,105	379 yes; 726 no
27-vertex higher-level control	1	LU yes; ordinary LC no

Table 1: Selected exact checks. Each row counts the objects named in that row; the counts do not form a single sample size.

These checks passed in a network-disabled container using Python and SymPy 1.14.0. The archive records the precise test families, scripts, outputs, exclusions, and mutation controls.

C Source versions and statement locators

Statement numbers in [4] refer here to the 27-page PDF of arXiv:2502.06566v3, rather than its HTML rendering or the 20-page proceedings version. Table 2 gives the HTML correspondence. Page numbers count from the first page of the PDF.

The integer normal-form complexity statement is Theorem 2 of Kannan–Bachem [8, p. 504]; the same page begins its full-row-rank rectangular extension. Our matrices have full column rank, so we apply that extension after transposition. The polynomial LU-decision question appears in Chapter 8, p. 107 of the v2 thesis [7]. The 27-vertex diagnostic is the graph pair described in Section 7, Figures 7–8, pp. 15–16 of arXiv:1611.06938 [3].

Result used	PDF statement	Page	HTML statement
Incidence and generalized operation	Defs. 1–2	4	Defs. 2.1–2.2
Clifford equations	Prop. 7	6–7	Prop. 3.1
Affine determinant solution set	Lemma 8	7	Lemma 3.2
Constrained decision	Prop. 10	8	Prop. 3.5
Local sets, types, standard form	Defs. 12–14	8–9	Defs. 3.8–3.10
Common standardization	Lemma 16	9	Lemma 3.12
Matching types and neighborhoods	Lemma 17	9	Lemma 3.13
One generalized operation	Lemma 18	10	Lemma 3.14
Complete incidence image	Lemma 19	10–11	Lemma 3.15
Auxiliary graph reduction	Lemma 20	11	Lemma 3.17
Auxiliary Clifford constraints	Lemma 21	11–12	Lemma 3.18
Logarithmic level suffices	Cor. 29	14	Cor. 3.29

Table 2: Statement correspondence in [4], arXiv v3. In that source, Appendices A, C, and D give the Clifford conversion, standardization, and auxiliary-graph reduction, respectively.

D The constrained Clifford test

We give the binary test used for the auxiliary graphs in Appendix A. Let Γ, Γ' be their adjacency matrices, of order N . Introduce $4N$ binary variables $\alpha_u, \beta_u, \gamma_u, \delta_u$ and their diagonal matrices A, B, C, D . The local-Clifford characterization [4, Proposition 7, pp. 6–7; Appendix A], originating in [1], is

$$\Gamma B \Gamma' + \Gamma A + D \Gamma' + C = 0, \quad (26)$$

$$\alpha_u \delta_u + \beta_u \gamma_u = 1 \quad (u \in V^\#). \quad (27)$$

Here $V^\#$ is the augmented vertex set, and all operations are in $\mathbb{F}_2$. The first equation is linear in the $4N$ variables: its (u, v) entry is

$$\sum_w \Gamma_{uw} \beta_w \Gamma'_{wv} + \Gamma_{uv} \alpha_v + \delta_u \Gamma'_{uv} + \mathbf{1}_{u=v} \gamma_u = 0. \quad (28)$$

The determinant conditions (27) are quadratic and cannot simply be omitted after Gaussian elimination.

The additional homogeneous linear equations for Lemma A.3 are those of [4, Lemma 21, pp. 11–12]:

$$\beta_u = 0 \quad (u \in Z), \quad \gamma_p = 0 \quad (p \in \bigcup_i P_i), \quad \beta_p = \beta_{p_i} \quad (p \in P_i \setminus \{p_i\}), \quad (29)$$

where p_i is one representative of each nonempty group. If $P = \bigcup_i P_i$, there are $\ell = |Z| + 2|P| - d \leq 2N$ such equations.

To see why these equations enforce the graph restrictions, use the conversion from a Clifford solution to a complementation sequence in [4, Appendix A]. At an auxiliary vertex, $\gamma_p = 0$ and (27) force $\alpha_p = \delta_p = 1$. If $\beta_p = 1$, complementing p on the source graph replaces β_p by $\beta_p + \delta_p = 0$, and leaves the other β coordinates unchanged. The auxiliary vertices are independent, so these operations may be performed first for all groups with $\beta = 1$. They leave identity local coordinates on the auxiliary vertices. The remaining solution has $\beta = 0$ outside W .

The coordinate updates are as follows. Complementing w on the source graph replaces its adjacency matrix by that of $G \star w$, and replaces the solution by

$$\begin{aligned} \alpha'_u &= \alpha_u + \mathbf{1}_{u=w} \gamma_w, & \beta'_u &= \beta_u + \mathbf{1}_{u=w} \delta_w, \\ \gamma'_u &= \gamma_u + \Gamma_{uw} \alpha_u, & \delta'_u &= \delta_u + \Gamma_{uw} \beta_u. \end{aligned} \quad (30)$$

These identities preserve (26)–(27) [4, Appendix A, pp. 18–20]. If some w has $\beta_w = \delta_w = 1$, this step removes w from the support of β and changes no other β coordinate. If β is nonzero but no such w exists, every vertex u in its support has $\delta_u = 0$ and $\gamma_u = 1$. The (u, u) entry of (26) then gives

$$\sum_w \Gamma_{uw} \beta_w \Gamma'_{wu} = 1. \quad (31)$$

Thus some neighbor w of u also has $\beta_w = 1$ and $\delta_w = 0$. Applying the three complementations u, w, u removes both u and w from the support of β , as follows by applying (30) three times; other β coordinates do not change. Repeating these reductions terminates with $\beta = 0$. The determinant equations then force $\alpha = \delta = 1$, and the diagonal part of (26) forces $\gamma = 0$. The remaining equation is $\Gamma = \Gamma'$, proving that the extracted sequence reaches the target.

Since every step above uses only vertices in the current support of β , the sequence remaining after the auxiliary steps is supported on W . No operation there changes an auxiliary neighborhood. Conversely, the same update identities, applied in reverse to a sequence of Lemma A.3, produce a solution of (26)–(29). An unused auxiliary has $\beta = \gamma = 0$; one used once has $\beta = 1, \gamma = 0$; and β stays zero on Z . Thus the solutions correspond exactly to the permitted sequences.

Lemma D.1 (Testing the determinant conditions). *On two connected graphs of order N , each with an even-degree vertex, equations (26)–(27) together with ℓ homogeneous linear constraints can be decided in $O((N^2 + \ell)N^2)$ binary operations.*

Proof. Compute a basis $b_1, \dots, b_k$ of the linear solution space S , with $k \leq 4N$, by Gaussian elimination. This costs $O((N^2 + \ell)N^2)$ operations. If $k \leq 4$, test all at most 16 vectors against (27).

For $k > 4$, the affine-structure result [4, Lemma 8, p. 7] states that the set $C \subseteq S$ additionally satisfying the determinant equations is empty or is an affine subspace of codimension at most two. Suppose $C = a + U$ is nonempty, with $\dim(S/U) = c \leq 2$. The images of the b_i span S/U ; select c of them as a quotient basis. The coset $a + U$ is a sum of at most c of those basis images. Hence one of

$$0, \quad b_i, \quad b_i + b_j \quad (i < j) \quad (32)$$

belongs to C . Test this list, of length $O(N^2)$, at a cost $O(N)$ per candidate. If none passes, C is empty. This is the finite search underlying [4, Proposition 10]. $\square$

Lemma A.2 establishes connectedness and the even-degree condition for both augmented graphs. Lemmas A.3 and D.1, with (29), therefore decide the required predicate in polynomial time in their actual order N .

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